PakEmis Sector Technical Note TN07

An Evidence-Gated IPPU Accounting Framework for Pakistan: Cement Material Flows and Admitted Process Emissions

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Abstract

Pakistan's industrial-process activity data are distributed across manufacturing statistics, trade records, company reports, product-use data and national greenhouse-gas reporting. We develop a reproducible Industrial Processes and Product Use (IPPU) framework that links each IPCC category to process-specific activity data and material balances. Categories lacking sufficient evidence remain explicitly unestimated, with their data requirements documented. The reference GHG module estimates cement calcination, ammonia production net of recovered carbon, iron and steel, and lubricant use. Its FY2023–24 total is 25.535 Mt CO_2; categories lacking supported methods and activity remain unestimated. The accompanying air product has its own admitted process coverage, including ammonia, steel and soda ash; this does not turn every IPPU GHG omission into an estimate.

A source-period audit shows that 48.011 Mt of cement previously attributed to FY2023–24 corresponds to FY2021–22; PBS and State Bank statistics both report 39.566 Mt for FY2023–24. Accounting for 3.792 Mt of clinker exports and 0.026 Mt of imports gives 33.440 Mt of inferred domestic clinker production. That Tier 1 inference gives a cement process estimate of 17.389–21.504 Mt CO_2 across the IPCC clinker fractions of 0.75 and 0.95, both ends priced as named national configurations. The reference is neither end: it is the clinker the producers themselves report.

All sixteen producer filings independently report 41.953 Mt of clinker, 125.5% of the low-clinker Tier 1 inference and 101.4% of the high case, which at the common factor gives the reference 21.816 Mt CO_2 (conditional 95% interval 20.227–23.452 Mt); the 0.75-fraction model is priced beside it as the national configuration *cement_tier1_model_low*. The gap between the census and the high-clinker inference indicates unresolved differences in reporting coverage, clinker transfers and stock movements rather than a small residual activity gap. A matched calendar-2022 reconstruction gives 17.985–22.595 Mt CO_2, compared with 27.906 Mt in Pakistan's official cement inventory. The full official IPPU inventory is 49.112 Mt CO_2e, including 19.400 Mt from chemical industries.

These results show that clinker trade and reporting-period alignment materially affect cement estimates, while company production data can potentially replace Tier 1 clinker assumptions. The national annual core now covers July 2015–June 2025. This note retains the detailed FY2023–24 clinker and reporting comparisons. Plant-specific ammonia recovery, nitric-acid abatement and fluorinated-gas equipment data remain consequential limitations.

Keywords: industrial processes, IPPU, clinker trade, material-flow accounting, F-gases, completeness, uncertainty, Pakistan

PakEmis Sector Technical Note TN07. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

Industrial Processes and Product Use (IPPU) is not one calculation. It includes carbonate decomposition in cement, lime and glass; feedstock carbon in ammonia and petrochemicals; nitrous oxide from nitric acid; carbon inputs to metals; non-energy uses of fuels; and fluorinated gases held in equipment and products (IPCC, 2006, 2019). These sources share an inventory sector but not an activity unit, emission factor or spatial representation. A credible national account must therefore do two things at once: connect each process to its own physical activity and identify explicitly which categories remain unestimated because process-specific activity data are insufficient.

Earlier work independently estimated cement calcination from one cement statistic, one assumed clinker fraction and one emission factor, but did not yet provide a sector-wide IPPU framework. It also omitted a required term in the cement method. When clinker production is inferred from cement, IPCC's Tier 1 worksheet requires clinker imports and exports because calcination emissions occur where clinker is produced rather than where it is ground or consumed. Pakistan is a substantial net clinker exporter, so that correction is material.

This study asks one question: *how far can a reproducible, physically coherent IPPU inventory be built from currently public Pakistani evidence while restricting emission estimates to categories for which the relevant process-specific activity data are available?*

We address three questions. First, what does a correct period- and trade-matched public-data calculation imply for cement calcination? Second, how closely does inferred national clinker agree with company-reported production, and which evidence still prevents a complete Tier 2 balance? Third, what does Pakistan's official reporting imply about the material categories that remain outside the independent estimate? We answer them by reconciling cement production and trade in a material balance, comparing inferred clinker with producer filings, auditing the CRT at cell level and assessing data completeness across 2.A–2.G.

The analysis combines a detailed FY2023–24 cement reconstruction with admitted ammonia, iron-and-steel and lubricant GHG estimates. The joint national release extends the supported annual methods through FY2024–25. Air species retain a distinct source/process coverage register, so an admitted process pollutant does not establish complete GHG coverage or the converse. Glass, lime, nitric acid and fluorinated gases retain their documented missing inputs.

2. Accounting Context

2.1. A heterogeneous inventory sector

IPCC Volume 3 organises IPPU into mineral industries (2.A), chemical industries (2.B), metal industries (2.C), non-energy products from fuels and solvents (2.D), electronics (2.E), fluorinated substitutes for ozone-depleting substances (2.F), and other product manufacture and use (2.G) (IPCC, 2006). The 2019 Refinement updates important methods and factors but does not remove the central data requirement: the activity must correspond to the physical process that emits.

This is why headline production quantities can be misleading. Cement output is not clinker production. Total fertiliser output differs from ammonia production. Billets do not identify the steelmaking route or its carbon inputs, and annual HFC leakage depends on the applications and equipment stocks in which the chemicals are used. Each quantity is valuable evidence, but the relevant process definition is also required.

2.2. Material balances as shared accounting

Several IPPU categories become clearer when written as conserved material flows. For cement, domestic clinker production plus imports minus exports and stock accumulation must equal the clinker ground into cement. For ammonia, feedstock carbon becomes gross process CO_2, which is divided among atmospheric emissions, urea production, other recovery and any permanent storage. For fluorinated gases, annual supply enters equipment banks and leaves through manufacturing losses, operating leakage, retirement, recovery and destruction.

These balances complement IPCC emission factors by constraining the activity bases to which those factors apply and expose residuals that a collection of independent equations can hide. They also connect IPPU to neighbouring sectors without double counting: cement kiln fuel remains Energy 1.A.2, while clinker calcination remains IPPU 2.A.1.

2.3. Evaluation and official reporting

Broad global products such as EDGAR are useful only after year, category, gas and GWP are matched. A national "Processes" total cannot validate cement alone. Pakistan's own CRT is a stronger accounting comparator because it exposes category rows, native gases, activity data, implied factors, recovery and notation keys (Government of Pakistan, 2026). We therefore use the CRT as an official method audit and scope comparator. Public production, trade and consumption datasets provide separate evidence for activity magnitude and reporting boundaries.

3. Data and Evidence Roles

3.1. Cement activity and its period audit

The official PBS Quantum Index of Manufacturing (QIM) reports 39.566 Mt of cement output for FY2023–24 (Pakistan Bureau of Statistics, 2024). The State Bank's FY2023–24 Statistical Supplement independently reproduces that value and provides a five-year fiscal series (State Bank of Pakistan, 2024): 39.121, 49.797, 48.011, 41.448 and 39.566 Mt for FY2019–20 through FY2023–24. This establishes that the 48.011 Mt value previously attributed to FY2023–24 belongs to FY2021–22. It is retained as a historical benchmark rather than a current structural alternative.

The annual core extends from FY2015–16 through FY2024–25 using held fiscal physical-production and trade records. The detailed reference-year analysis below remains FY2023–24. Publication revisions and source vintages are

retained rather than interpreted as competing observations for one unchanged boundary.

QIM also reports FY2023–24 production of soda ash (0.785 Mt), billets and ingots (4.914 Mt), glass plates and sheets (20.205×10^6 m^2) and lubricating oil (130.513×10^6 L). These statistics establish the occurrence and scale of the activities but do not provide every variable required by IPCC. Glass needs mass, type, cullet and carbonate inputs; steel needs the production route and carbon inputs; lubricants require non-energy consumption and oxidation rather than output.

Monthly UN Comtrade records for HS 252310 were summed from July 2023 through June 2024 (United Nations Statistics Division, 2024). Exports total 3.792 074 Mt and imports 0.026 131 Mt, with all twelve months present for both flows.

The panel now extends back to July 2014 for both flows, and every fiscal year of the national series resolves to an observed trade term. Extending it required deciding what an EMPTY month means, because the source returns no record both for a month in which nothing was traded and for a month it has not processed, and the response does not distinguish them. Pakistan's clinker imports are small and sporadic—0.026 131 Mt across the whole of FY2023–24 against 3.792 074 Mt exported—so empty months are common, and reading them as zeros would manufacture a complete panel from an incomplete one, in the direction that lowers inferred domestic production. Each calendar year's annual-frequency total was therefore retrieved as a control: a month with no record counts as zero only where the monthly records of its calendar year already sum to that year's annual total, so the source itself certifies that nothing further moved. Where they do not sum, the month is unobtained and the fiscal years touching it are NE. Months lacking a record count as zero only after this compatible source-control check; unresolved coverage remains explicit. The per-year verdicts are generated by `pkinv.sectors.ippu.trade_evidence`. For the official comparison year, January–June 2022 QIM observations were joined to July–December 2022 and sum to 44.331 Mt. Calendar-2022 WITS/UN Comtrade records give 1.341 580 Mt of exports and 0.003 999 Mt of imports (World Bank, 2022).

3.2. *Producer and facility evidence*

We identified 16 cement producers relevant to FY2023–24. All sixteen public filings disclose 41.953 170 Mt of clinker production (Lucky Cement Limited, 2024; D.G. Khan Cement Company Limited, 2024; Attock Cement Pakistan Limited, 2024; Power Cement Limited, 2024; Cherat Cement Company Limited, 2024; Pioneer Cement Limited, 2024; Thatta Cement Company Limited, 2024; Bestway Cement Limited, 2024; Kohat Cement Company Limited, 2024; Dandot Cement Company Limited, 2024; Gharibwal Cement Limited, 2024; Dewan Cement Limited, 2025; Fauji Cement Company Limited, 2024; Maple Leaf Cement Factory Limited, 2024; Flying Cement Company Limited, 2024). Where disclosed, we also compiled opening and closing stocks, clinker purchases and imports, clinker ground, domestic sales and exports. Missing flow terms remain unobserved, so company balances are evaluated only where the necessary data are available.

Flying's scanned annual report gives exact FY2023–24 clinker and cement production of 0.322 624 Mt and 0.321 500 Mt, respectively; Fecto's FY2025 annual report states the FY2023–24 comparative directly, 0.651 048 Mt of clinker and 0.724 209 Mt of cement, where its earlier public briefing had given only capacity and rounded utilisation (Fecto Cement Limited, 2024, 2025; Flying Cement Company Limited, 2024). All 16 census companies therefore report an exact clinker quantity; what no company reports is the balance around it (purchases, sales, grinding and stocks). The census is `ippu_cement_producer_census_2024.csv`.

The facility registry identifies cement and fertiliser installations by operator, process route and administrative area. Coordinate provenance is retained for each record. The primary geographic package uses only locations compatible with its declared licence, while ODbL-derived locations retain a separate alternative. Company-wide clinker is split conditionally among works where plant-specific output is unavailable; unsupported placement remains unallocated. A point location does not itself establish plant production.

3.3. *Official reporting, chemicals and product-use data*

Pakistan's 2024 CRT submission contains a dedicated workbook for inventory year 2022 (Government of Pakistan, 2026). We preserve ZIP and workbook hashes, worksheet names, cell addresses, raw values and formulas. Table 2(I) provides category emissions; Table 2(I).A–H provides activity, implied factors, native gases and recovery; and Table 2(II).B–Hs2 provides species-level F-gas records.

Public company reports provide initial evidence for the next categories. Fatima reports 0.311 242 Mt of calendar-2024 ammonia production at Sheikhupura; historical records identify ammonia and nitric-acid output at Sadiqabad, dual-pressure nitric-acid technology and CDM abatement projects at Sadiqabad and Pakarab (Fatima Fertilizer Company Limited, 2024; UNFCCC, 2006, 2011). The NFDC annual review adds four extracted tables: domestic fertiliser production by product and manufacturer, product-wise offtake split domestic and imported, plant-wise fertiliser-sector gas, and the same gas monthly by plant for both fiscal years. The feedstock/fuel split is therefore observed for the reference year itself— 83.5 %/16.5 % in 2023–24, and 83.2 %/16.8 % in 2024–25—rather than borrowed backwards from the later year. The reference year's total, 751.9 MMCFD, agrees with the regulator's independently published 756 to 0.5 %. These support a four-pathway activity reconciliation for ammonia—official filing, plant census, product stoichiometry and gas balance—whose pathways do not yet agree. National plant coverage, recovered CO_2, and matched abatement utilisation remain incomplete, so these observations support the category assessment but are not used to estimate national emissions. Calendar-2022 and calendar-2024 records are audited separately. Verified CDM reductions are retained by monitoring period and are not carried forward as current N_2O emissions or abatement rates. For Pakarab, the

period-specific audit preserves 2008 line production, operating days, nameplate capacity, tertiary De-N_2O technology and the approved baseline cap without turning those historical observations into FY2023–24 activity.

Pakistan's Ozone Secretariat profile reports HFC consumption of 9.815 266 Mt CO_2e in 2024 (Ozone Secretariat, 2026); Pakistan ratified the Kigali Amendment on 22 October 2025 as an Article 5 Group 2 party, whose baseline is the 2024–2026 consumption average, and the profile records that no HFC licensing system is yet established. Consumption is not annual leakage. Estimation therefore requires annual data by species and application on imports, exports, production, destruction, new and servicing charge, equipment retirement and recovery, together with the parameters needed by the equipment-bank model. Two free instruments sit beside the consumption anchor: UN Comtrade's HS-2022 HFC lines (net imports of 6.0 kt of substance in 2024, bulk plus mixtures — a customs corroboration of supply that cannot resolve species) and a one-stratum CO_2e-weighted bank model run on the consumption series itself under declared IPCC default losses, whose 2024 reading of 4.42 Mt (2.3–6.2 across the 5–15 % loss-rate cases, not a bound) ships as the screening configuration `scope_plus_fgas_bounded_bank`, outside every reference total.

4. Methods

4.1. Category completeness and common calculation structure

Each detailed IPPU category is assigned an IPCC notation key together with its available activity data, methodological status, source provenance, materiality and remaining data requirement. Emissions are calculated only when both the activity datum and the process definition satisfy the relevant IPCC method. A production statistic may therefore establish that an activity occurs without being sufficient to quantify its emissions. This distinction keeps unavailable categories separate from reported zeros and excludes unrelated proxy data from the national total.

All estimated categories use a common calculation structure containing the IPCC code, activity value and unit, emission factor, source, method and uncertainty. Facility-level data can replace aggregate national activity when they become available without changing the underlying accounting method. The facility inventory receives emissions only when activity is reported for the same plant boundary.

4.2. Cement clinker balance

For a fiscal year ending y, clinker contained in cement is

$$C_y^{\text{cement}} = P_y^{\text{cement}} f^{\text{clinker}}, \tag{1}$$

and domestic clinker production is solved as

$$C_y^{\text{production}} = C_y^{\text{cement}} - I_y + X_y + \Delta S_y, \tag{2}$$

where I and X are clinker imports and exports and ΔS is the stock increase. This is the sign convention implied by the IPCC Tier 1 worksheet: production plus imports minus exports and stock increase equals clinker ground into cement (IPCC, 2006). No public stock series was identified, so $\Delta S = 0$ in the reference configuration and stock change remains an explicit data gap.

Process CO_2 is

$$E_{CO_2,y} = C_y^{\text{production}} \, \text{EF}_{\text{clinker}}, \tag{3}$$

with an IPCC default factor of 0.52 t CO_2 per tonne of clinker. The estimate covers calcination only. Kiln fuel belongs to Energy 1.A.2 and shares the facility account while remaining outside 2.A.1. The air calculation uses an integrated cement-stack factor once, rather than adding fuel and process factors for that same stack.

4.3. Structural and parametric uncertainty

The source audit removes the apparent activity-source choice for FY2023–24: QIM and SBP agree on 39.566 Mt, while 48.011 Mt belongs to FY2021–22. Observed clinker trade is required by Equation 2 and is included in every valid current model. The remaining declared structural choice is the clinker fraction, evaluated at 0.75 and 0.95. Two no-trade calculations are retained separately as historical method benchmarks: one uses the correct FY2023–24 activity to isolate trade omission, and the original result combines trade omission with the superseded period tag. Neither enters the valid structural range.

Within both valid configurations, a seeded 10,000-iteration Monte Carlo samples bounded triangular distributions for cement activity (±5%), imports and exports (±10%), and the clinker factor (±8%). The clinker fraction is fixed within each structural configuration. We report conditional P2.5, median and P97.5 values rather than merging discrete source/method choices into one probability interval.

For calendar 2022, the same equation is evaluated with twelve QIM monthly observations and calendar-year clinker imports and exports. This makes the comparison with Pakistan's 2022 CRT period-matched; the clinker fraction and zero stock change remain structural assumptions.

4.4. Official-reporting audit

The CRT audit extracts the native category cells rather than transcribing the published total. For every numeric background row and gas, final emissions are reconstructed as

$$E^{\text{final}} = A \times \text{IEF} + R, \tag{4}$$

where recovery R is reported as a negative quantity. This treatment is important for ammonia, where gross process CO_2 is partly recovered for downstream use. Arithmetic closure tests the workbook; comparison with public evidence tests its activity, factor and boundary choices.

4.5. Spatial and temporal products

The current cement provincial allocation preserves the legacy cement/limestone proxy only so that downstream national and province tables remain reconciled. Its former Punjab activity anchor is no longer treated as period-matched.

Category	FY2023–24 status	Available public evidence	Evidence still required
2.A.1 Cement	E	QIM, trade, 16 producer filings	common boundary, plant splits, clinker movements, stocks and chemistry
2.A.2 Lime	NE	Mineral/industry screen	Marketed and captive lime by type
2.A.3 Glass	NE	QIM area output	Mass, glass type, cullet and carbonate inputs
2.B.1 Ammonia	E	NFDC production by manufacturer, plant gas feed/fuel split, one FY2024 plant, CRT/history	Plant-level output and recovered CO_2 for the Tier 2 step
2.B.2 Nitric acid	NE	Plant and CDM evidence	FY2024 100%-acid output and abatement operation
2.C.1 Iron/steel	E	QIM billets/ingots	Process route and carbon inputs
2.D.1 Lubricants	E	QIM output	Non-energy use and oxidation
2.F HFCs	NE	Ozone consumption	Species/application flows and equipment banks
2.G SF_6/other	NE	Occurrence screen	Equipment banks, refills, recovery and use data

Table 1: Evidence status for the principal IPPU categories. The complete supplementary matrix covers every detailed 2.A–2.G category and documents the data needed for estimation.

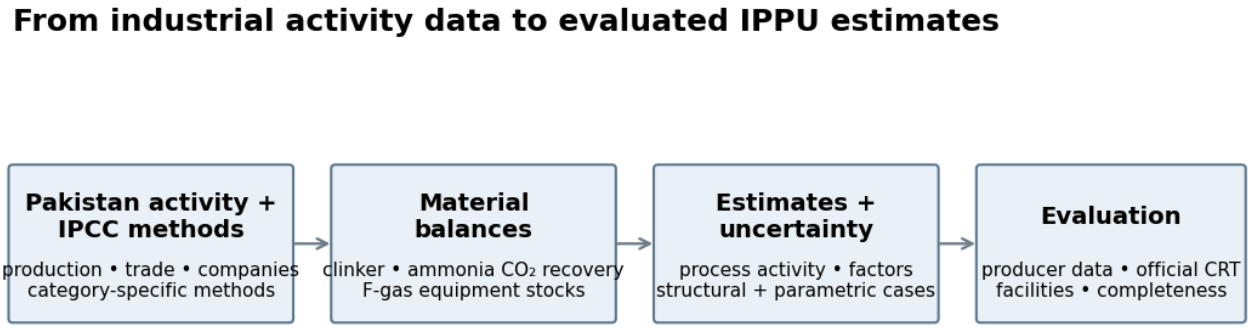


Figure 1: Public inputs enter conserved material balances before category emissions are calculated. Categories with insufficient evidence retain their notation keys and documented data requirements.

Industrial process emissions are point sources, so the target product is a facility ledger that reconciles plant, district, province and national totals. Facility records retain their source and coordinate licence. Only locations compatible with the primary package enter its facility-assisted allocation. ODbL-derived alternatives retain separate identities and obligations. Company-wide output is distributed among its works conditionally when plant-specific production is unavailable. Unlocated or excluded activity remains in the unallocated account; it does not establish native facility resolution.

5. Category Methods and Data Requirements

The framework defines an IPCC calculation method and an evidence requirement for each category. Higher-resolution activity data can therefore replace aggregate statistics without changing the material balance or category definition.

Mineral industries. Cement uses Equation 2. Generic carbonate methods are specified for lime, glass and other carbonate uses, but their Pakistan rows remain NE until lime output by type, glass mass and type, cullet use and carbonate inputs are compiled.

Chemical industries. The ammonia ledger divides gross process CO_2 among atmospheric emissions, recovery into urea, other recovery and permanent storage and asserts zero residual. Nitric-acid emissions are estimated from production expressed as 100% acid, technology-specific N_2O factors, the abated share, destruction efficiency and abatement utilisation. Total fertiliser output or nameplate capacity alone is insufficient for either estimate.

Metals and non-energy products. QIM reports billet, ingot and lubricant production. Iron and steel estimation additionally requires the production route and carbon inputs; lubricant emissions require non-energy consumption and oxidation. These remaining inputs are documented in the completeness matrix.

Fluorinated gases. A species-resolved equipment-bank model conserves annual chemical supply across manufacturing loss, installed bank, operating leakage, retirement, recovery, destruction and recovered gas available for reuse. AR5 and AR6 GWP registries include the HFC, PFC, SF_6 and NF_3 species required by the CRT. The Ozone profile lacks the species- and application-level data needed to estimate the bank, so 2.F remains NE in the independent FY2023–24 inventory.

6. Evaluation Design

Each evidence layer addresses a distinct question (Table 2).

Automated quality-control tests verify clinker-balance closure, completeness of monthly trade records, correct trade signs, inclusion of observed trade in current estimates, reproduction of reported CRT activity–factor calculations and full 2.A–2.G coverage in the completeness matrix. Provincial allocations must also reconcile exactly to the national total. Detailed test definitions are provided in Appendix A.

The official comparison is category-specific. Pakistan's 2022 cement row is compared with an independent calendar-2022 QIM/trade reconstruction under both clinker-fraction choices. The official whole-IPPU total is not compared with the independently estimated cement component; its category composition instead establishes the materiality of categories still missing from the public-data estimate. Company clinker is an independent activity test, but not yet a national Tier 2 result: every one of the sixteen producers now reports an exact clinker quantity, but producer coverage, transfers and stocks have not been reconciled with QIM cement production.

7. Results

7.1. The activity audit changes the reference calculation

The apparent discrepancy in FY2023–24 cement activity results from an incorrect period assignment. PBS QIM and the SBP fiscal series both report 39.566 Mt; the former 48.011 Mt statistic is the FY2021–22 value. The corrected reference therefore contains 29.6745 Mt of clinker at a 0.75 fraction. Adding 3.792 074 Mt of exports and subtracting 0.026 131 Mt of imports gives 33.440 443 Mt of inferred domestic clinker production (Table 3). At 0.52 t t^{-1}, process emissions on that inferred route are 17.389 030 Mt CO_2—the named Tier 1 diagnostic. The reference does not use the inferred route: it takes the 41.953 170 Mt of clinker the sixteen producers report and gives 21.815 648 Mt CO_2 at the same factor.

Using the correct FY2023–24 activity but omitting observed trade gives 15.430 740 Mt CO_2. Trade correction therefore adds 1.958 Mt, or 12.7%, and is a method correction rather than a valid structural alternative. The original 18.724 Mt result is retained only as a historical benchmark because it combines trade omission with the misdated 48.011 Mt activity.

7.2. Producer evidence against the two Tier 1 inferences

All sixteen company filings sum to 41.953 170 Mt of actual clinker, equivalent to 21.815 648 Mt CO_2 at the common factor—the reference activity. This observed census is 125.5% of the 0.75 Tier 1 inference and 101.4% of the 0.95 inference—a near-identity that partly reflects the two constructions drawing on overlapping reporting, and is therefore construction similarity as much as corroboration; Fecto's quantity is read from its FY2025 annual report's comparative table (Figure 2). We use the reported census without extrapolation. If the company filings and QIM cover the same producers and reporting period, and stock change is negligible, the observed subset implies a diagnostic clinker fraction of 0.949. The low-clinker case is therefore inconsistent with the company data unless the two sources differ in producer coverage or reporting period, or unless clinker transfers and stock changes materially affect the balance.

The company compilation constrains the national clinker balance but does not yet constitute a complete Tier 2 activity dataset. A consistent set of opening and closing stocks, clinker purchases, domestic sales, exports and clinker ground has not been reported. The unresolved terms are documented without allocating the national residual among firms.

Comparison	Purpose	Independence/limit
Clinker material closure	Physical accounting	Depends on cement fraction and zero stock change
QIM versus SBP fiscal series	Period and source audit	Independent official publications agree on FY2023–24
Calendar-2022 cement versus CRT	Official method comparison	Matched year and category; clinker fraction remains assumed
Producer clinker versus Tier 1	Independent activity/boundary evaluation	All sixteen producers report clinker; movements and stocks incomplete
CRT arithmetic reproduction	Spreadsheet QA	Tests arithmetic, not methodological validity
Ozone HFC consumption versus CRT	Completeness diagnostic	Consumption and annual emissions have different boundaries
Provincial proxy closure	Allocation QA	Reconciles exactly but is not spatial validation

Table 2: Evaluation roles. Agreement is interpreted only at the boundary supported by each comparison.

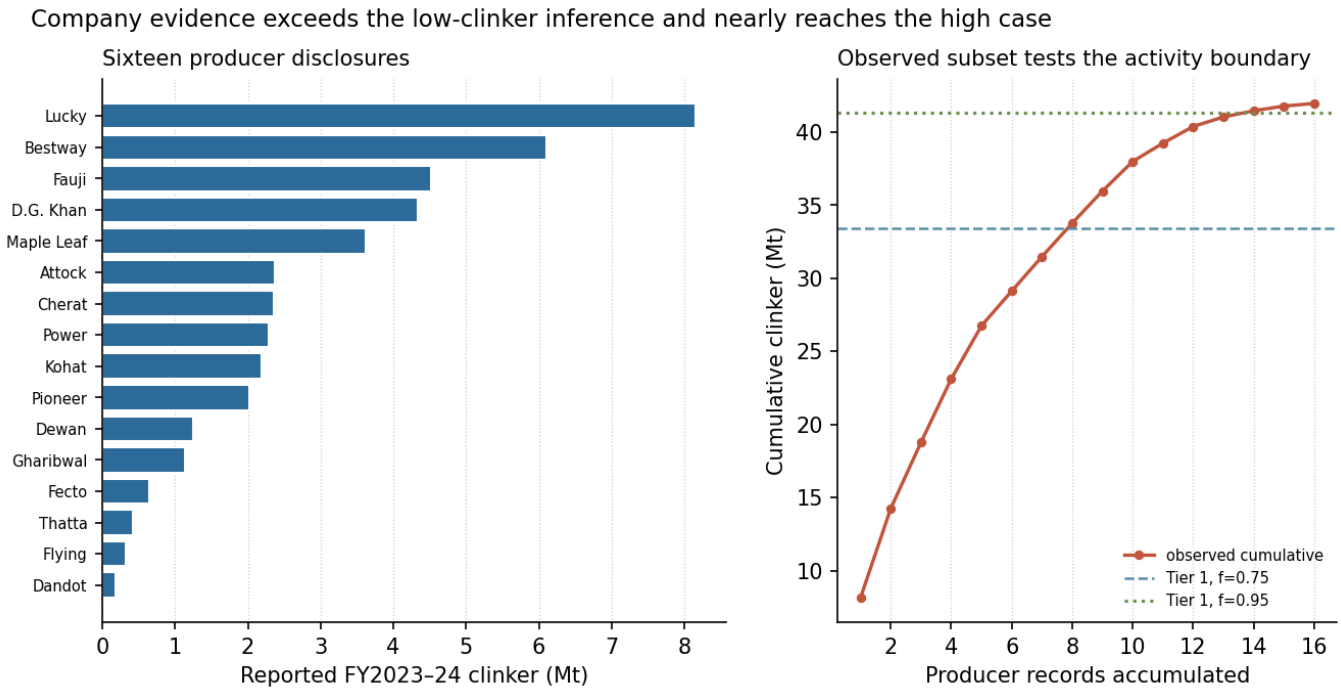


Figure 2: Reported FY2023–24 clinker by producer (left) and the cumulative observed subset against the two trade-corrected Tier 1 inferences (right). All sixteen census producers report an exact quantity; Fecto's is the FY2024 comparative of its FY2025 annual report.

Clinker flow	Mt	Sign in inferred production
Clinker contained in cement	29.674	+
Imports	0.026	–
Exports	3.792	+
Stock increase	0.000	+
Inferred domestic production	**33.440**	

Table 3: FY2023–24 clinker balance. The zero stock row is explicit and unresolved.

7.3. Valid cement models and conditional uncertainty

With observed trade included in every current model, the structural range is 17.389–21.504 Mt CO_2 for clinker fractions of 0.75 and 0.95 (Figure 3); both ends of that range are priced against the national total as named configurations in `outputs/national_configurations.csv`, and the observed census (21.816 Mt) sits 1.4% above the high end. The reference Monte Carlo has a median of 21.809 297 Mt and a 95% interval of 20.227–23.452 Mt. The all-Portland configuration has a median of 21.493 Mt and an interval of 19.937–23.062 Mt. The method benchmarks are shown separately; the valid range is based exclusively on trade-corrected models.

7.4. Calendar-2022 reconstruction sharpens the official comparison

The 12 monthly QIM observations sum to 44.331 Mt in calendar 2022. Combining them with 1.341 580 Mt of clinker exports and 0.003 999 Mt of imports gives 34.585 831 Mt of inferred clinker and 17.984 632 Mt CO_2 at the 0.75 fraction. The 0.95 configuration gives 22.595 056 Mt.

Pakistan's CRT cement row reports 50.2455 Mt of activity and an implied factor of 0.55539 t/t, yielding 27.905 848 Mt CO_2. The independent matched-year reconstructions are therefore 35.6% and 19.0% below the official result. The gap reflects both activity (44.331 versus 50.246 Mt) and clinker/factor treatment; it is no longer an artefact of comparing different years.

The CRT reports 49.112 03 Mt CO_2e for all IPPU in 2022. Mineral industries account for 29.300 Mt and chemical industries 19.400 Mt; non-energy products and other industrial processes contribute 0.217 and 0.192 Mt. All seven numeric background calculations reproduce exactly, including the ammonia recovery term. The official discrepancy is therefore methodological and empirical rather than spreadsheet arithmetic.

7.5. Evidence available for the next categories

Cement, ammonia, iron and steel, and lubricant use enter the FY2023–24 GHG module. Cement contributes 21.816 Mt CO_2 to the 25.535 Mt module total; the following plant evidence identifies remaining process and Tier 2 gaps rather than implying ammonia is wholly absent. One ammonia plant reports 0.311 242 Mt of output in 2024, and historical plant and CDM records identify nitric-acid technology and abatement projects. National estimates require complete plant coverage, recovered CO_2 and annual abatement operation. For ammonia the generated reconciliation now compares four independent activity pathways and locates the inconsistency: the official 2022 filing's 11 467 kt of ammonia would require roughly half again more gas than the whole fertiliser sector draws, embodied ammonia in NFDC's product-wise production sits near 4500 kt, and the quantifiable plant floor is 548 kt from two of eight census plants. The filing's recovery term, by contrast, implies 6442 kt of urea, consistent with audited production; the suspect leg is the activity, not the recovery. The HFC evidence similarly requires species- and application-specific equipment data before consumption can be related to annual emissions.

Chemical observations are reported separately by year, and verified CDM reductions are assigned only to their stated monitoring periods. For Pakarab, the available evidence includes line-level production, operating days and abatement technology for 2008.

Cement has the most detailed company-level production support. Ammonia retains product-derived output and recovery assumptions; nitric acid still lacks complete own-year production and demonstrated abatement operation. The primary geographic product preserves company-level uncertainty and licence-compatible locations, with unsupported placement retained in the unallocated account.

7.6. Air-pollutant process coverage

The admitted air classes estimate ammonia-process releases, electric-route steel emissions and soda-ash process emissions with their specific factors and physical activities. Cement air factors cover an integrated kiln stack and are accounted once under industry; cement GHG calcination remains separate from kiln fuel. Urea finishing, nitric-acid pollutants, lime, glass and other unresolved processes remain explicit coverage gaps. Missing species are not filled with zero or with an uncontrolled screening endpoint.

8. Discussion

8.1. Trade correction is part of the method

Pakistan's net clinker exports are 12.7% of the correctly period-tagged no-trade estimate after conversion to CO_2. Omitting them assigns calcination to the country where exported clinker is ground rather than the country where it was produced. The corrected equation therefore changes both the magnitude and the geographic boundary of 2.A.1. Stock movement remains unresolved, but it is now visible as one term in the same physical balance.

The source audit also illustrates why period and source documentation are part of the calculation. The reported 48.011 and 39.566 Mt values were consecutive observations in the same official fiscal series, rather than competing estimates for one year. Correcting the period changes the headline result more than the remaining parametric uncertainty and removes a spurious structural choice.

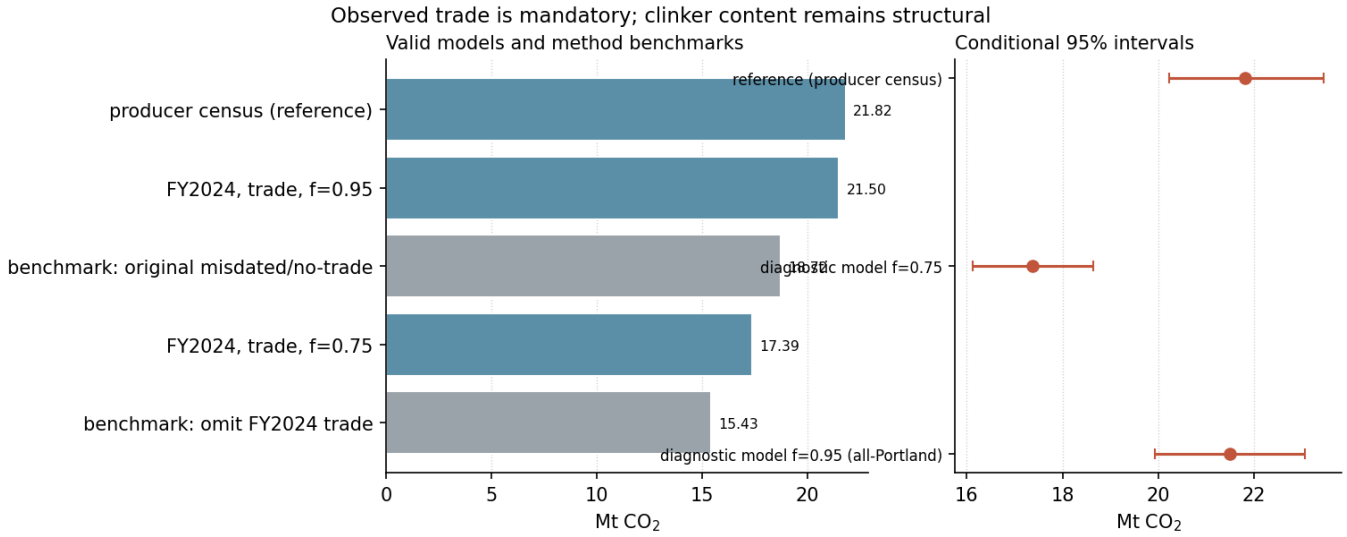


Figure 3: Left: valid trade-corrected clinker-fraction models and separate method benchmarks. Right: conditional parametric intervals within the two valid configurations.

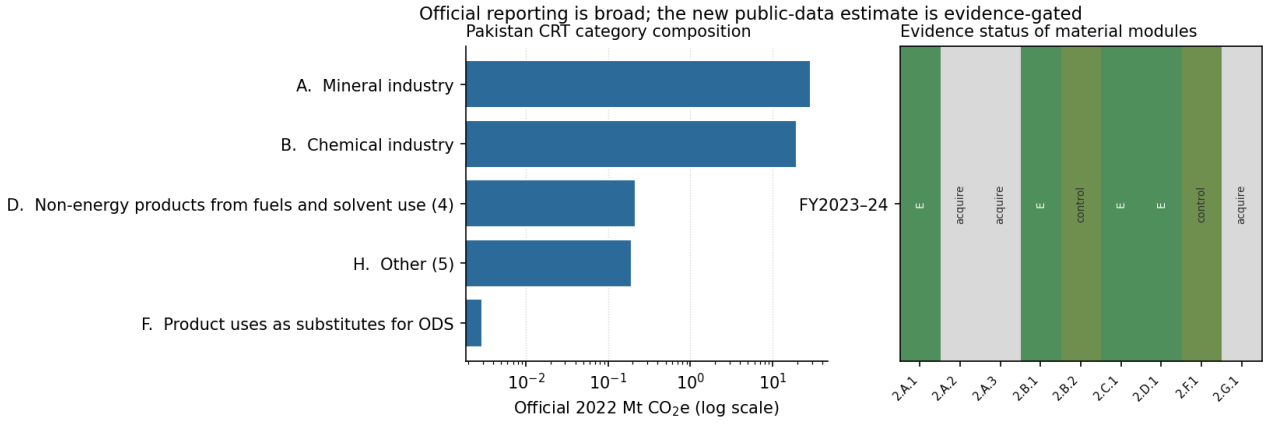


Figure 4: Official 2022 IPPU category composition (left) and independent FY2023–24 evidence status (right). Cement, ammonia, iron and steel and lubricant use are estimated; unsupported rows remain NE.

8.2. A complete sector starts with completeness accounting

Pakistan's official results make the scope problem concrete. In 2022, chemical-industry emissions were two-thirds as large as mineral-industry emissions. A cement calculation can therefore be rigorous without representing the full IPPU sector. The framework avoids conflating the two by assessing the evidence and method available for every 2.A–2.G category before calculating a sector total.

This design also changes how public data are interpreted. A soda-ash production record is useful for 2.B.7 only after the production route is known and must be distinguished from soda-ash consumption under mineral uses. Total fertiliser output cannot estimate ammonia, where process CO_2 recovery into urea is central. Billets identify steel output but not the production route or carbon inputs. HFC consumption establishes chemical supply, whereas annual emissions depend on equipment stocks and leakage. These distinctions preserve the physical meaning of each category estimate.

8.3. The CRT audit identifies immediate empirical priorities

All tested CRT rows reproduce exactly from their reported activities, implied factors and recovery terms. Its 49.112 Mt result is consequently traceable to reported activities, factors, recovery and omissions rather than a spreadsheet mistake. Three empirical priorities follow.

First, the sixteen-company clinker census—complete for production since Fecto's exact output was read from its annual report—should be completed with clinker purchases, sales and stocks. The census already exceeds the low-clinker inference and slightly exceeds the high case, so these data must reconcile reporting boundaries as well as replace Tier 1 activity. Second, plant ammonia and nitric-acid records can reconstruct the official chemical-industry result while preserving recovered carbon and abatement. Third, species- and application-level HFC imports, exports and equipment stocks can support an estimate based on the Ozone consumption data. The small HFC value in the 2022 CRT and the much larger consumption statistic have different accounting boundaries, but together they make product-use completeness a high-priority audit. The CRT entry is smaller than "small" conveys and its shape matters: all of category 2.F is a single species, HFC-23, entered at 0.236 955 t as exactly 1 % of a declared 23.695 49 t operating stock and filed under sub-category 2.F.1.a commercial refrigeration, with every other species and sub-category `NE` or `NO`. HFC-23 is an HCFC-22 by-product and a cascade low-temperature refrigerant rather than a commercial refrigerant, so the species assignment is a reporting artefact, and Pakistan's first Biennial Transparency Report then books 2.F, 2.C, 2.E and 2.G all at zero. The two most recent official submissions therefore disagree about whether the category exists, which is not a boundary difference.

The leak rate is the other decision that is easy to overlook, and `outputs/pakemis_fgas_leak_rate_adjudication.csv` prices it. The species-resolved bank applies one operating loss fraction per stratum—refrigeration 8 % per year, stationary air-conditioning 6 %, mobile air-conditioning 15 %—while IPCC Table 7.9 publishes a rate for each sub-application. Mobile air-conditioning maps one-to-one onto its table row and needs no weighting. Stationary air-conditioning lumps chillers with residential and commercial units, whose bands overlap, and 6 % is admissible for either part. Refrigeration lumps five sub-applications whose operation-emission bands span 0.1 % to 50 % per year with no common value, and the single 8 % lies outside two of them, from 16 to 80 times above domestic refrigeration's whole band. For that part the rate is not uncertain but inadmissible, and the split that would resolve it is not held.

What the envelope shows is that the rate decides timing rather than mass. Re-running the bank across refrigeration's full band moves the operating channel from 0.013 Mt to 2.062 Mt of CO_2e, a factor of one hundred and fifty-five, while the stratum total moves only 0.757 Mt to 2.192 Mt: gas the operating channel does not leak is still emitted at retirement, because no recovery is credited anywhere. The deferred mass appears instead as a closing bank of 1.959 Mt to 12.559 Mt of CO_2e. A low leak rate is therefore not a smaller inventory but a larger future liability, and category 2.F contributes zero to every reported total throughout.

8.4. Facility geography is preferable to proxy geography

The current provincial cement allocation is retained for compatibility with the national inventory, but its earlier Punjab activity anchor is not period-matched. Cement kilns, ammonia plants, nitric-acid plants, soda-ash works, steel mills and glass plants are point sources. A shared facility inventory can assign process emissions to their physical sites and link the same installation to Energy 1.A.2 combustion without mixing the two categories. Diffuse product use, especially refrigeration and air-conditioning, can then be represented through equipment stocks and population or service activity rather than an industrial point proxy.

8.5. The matched official comparison reveals a second cement discrepancy

Calendar matching does not close the official cement row. Even the 0.95 public-data configuration remains 19% below the CRT, while the low-clinker model is 36% below. Part of the difference is visible in the activity data: the CRT uses 50.246 Mt against 44.331 Mt from the monthly QIM reconstruction. The remainder is embedded in the CRT's 0.55539 t/t implied factor and any clinker, stock or coverage treatment behind it. Because the workbook arithmetic closes, resolving this difference requires the institutional derivation and company clinker balance rather than further arithmetic verification.

8.6. Transferability

The framework is transferable because its accounting structure is independent of any one national data source. Countries with incomplete industrial statistics commonly combine

national production totals, trade data, partial company reporting, ozone-substitute consumption records and official inventory submissions. Higher-quality evidence can replace aggregate activity assumptions without changing the material balances or category definitions. Exact mass closure, category completeness and an auditable official crosswalk remain valid as the quality of the underlying evidence improves.

9. Limitations

The independent FY2023–24 estimate covers four IPPU categories: cement calcination (2.A.1, 21.815 648 Mt), ammonia production (2.B.1, 3.2433 Mt), iron and steel (2.C.1, 0.3931 Mt) and lubricant use (2.D.1, 0.0833 Mt), for a module total of 25.5354 Mt of CO_2. Methods and data requirements are defined for the categories that remain unestimated — lime (2.A.2), glass (2.A.3), nitric acid (2.B.2), HFCs (2.F) and SF_6 (2.G) — and those stay NE until the required activity and process data are available. Cement is therefore 85.4 % of the module rather than all of it: the 21.815 648 Mt figure represents cement-process CO_2 rather than total IPPU or total cement emissions, and the 17.389 Mt Tier 1 inference at the 0.75 clinker fraction is a named diagnostic beside it, not the reference.

National clinker production is observed at the company grain; stock change is not. All sixteen producer records together exceed the low-clinker inference by a quarter and the high case by 1.4%, but the company-level purchase, sale, grinding, export and stock fields remain missing. The company compilation therefore constrains the national clinker balance but does not yet constitute a complete Tier 2 activity dataset.

The conditional Monte Carlo covers activity, trade and factor ranges within each clinker- fraction model. It does not assign a probability to the 0.75 or 0.95 configuration. Stock uncertainty is not sampled because no defensible public bound was identified.

The matched calendar-2022 comparison still depends on an assumed clinker fraction and zero stock change. It demonstrates that year mismatch does not explain the difference from the CRT, but it cannot reconstruct the official institutional derivation without its internal clinker and coverage evidence.

The chemical plant table is partial. Nameplate capacity, one calendar-2024 ammonia observation and CDM project existence cannot substitute for national plant production, recovered CO_2 or annual abatement operation. Ozone-profile HFC consumption provides a national supply statistic but cannot be compared directly with annual emissions.

Company-level clinker production does not establish plant-level output. Facility-assisted geography uses only the source locations compatible with the primary package's licence; ODbL-derived alternatives remain separate. Unsupported or excluded locations remain unallocated rather than inheriting another source's coordinates.

The common annual core covers ten fiscal years, July 2015–June 2025. The reference-year company analysis does not imply observed annual plant geography. Soda ash production is observed; its Solvay-route GHG treatment places the coke combustion term under Energy, where missing coke activity remains explicit. The IPPU notation must not be read as evidence that soda ash production does not occur. The admitted soda-ash air process class has its own factor and boundary. Lime, glass, nitric acid and fluorinated-gas emissions remain unresolved where required process or equipment data are missing.

10. Conclusion

This study establishes an IPPU-wide accounting framework for Pakistan while independently estimating cement calcination, ammonia production, iron and steel, and lubricant use within documented boundaries. Every 2.A–2.G category is assessed, industrial material balances are conserved and official reporting is extracted at cell level. Categories lacking sufficient evidence remain explicitly unestimated, with their remaining data requirements identified.

Three results shape the cement estimate. The source audit establishes that 48.011 Mt is FY2021–22 cement output and that the FY2023–24 reference is 39.566 Mt. Applying observed clinker trade to that activity gives 17.389 Mt CO_2 at a 0.75 clinker fraction, compared with 15.431 Mt when trade is incorrectly omitted. The Tier 1 configurations span 17.389–21.504 Mt across the two IPCC clinker-fraction cases—0.75 at the low end, 0.95 at the high—and the span is a structural benchmark range, not a number with a preferred point. The reference lies outside it: it is the producer census's own disclosed clinker, 21.816 Mt CO_2 (conditional 95% interval 20.227–23.452 Mt), 1.4% above the high case, and the 0.75-fraction model is priced beside it as the *cement_tier1_model_low* configuration. The model span remains distinct from the producer-based reference.

All sixteen producer filings now disclose actual clinker—41.953 Mt, 125.5% of the low-clinker model inference and 101.4% of the high case, with Fecto's 651 048 t read from its FY2025 annual report's comparative table rather than imputed from capacity—and that census is the reference activity, with the Tier 1 model beside it as the named diagnostic. What remains for the movement ledger is the material balance: stocks, transfers and clinker ground, the Tier-2 fields no filing publishes.

The matched calendar-2022 reconstruction gives 17.985–22.595 Mt, below the official cement row of 27.906 Mt. Pakistan's full CRT reports 49.112 Mt CO_2e, including 19.400 Mt from chemicals. The next empirical priorities are consequently bounded: finish clinker production and movements; complete plant ammonia/nitric-acid production, recovery and abatement; and compile the species- and application-level data required for an HFC equipment-bank model.

The current estimates cover cement calcination, ammonia production, iron and steel and lubricant use; the same framework provides defined pathways for nitric-acid and fluorinated-gas estimates as the required plant- and application-level data become available.

Code and data availability

The calculation code, activity data and generated outputs are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors) and reproduce from the commands in Appendix A. Numerical results are read from generated CSV files, with automated checks confirming agreement between the manuscript and calculations.

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Appendix A. Reproduction and generated outputs

```
pip install -e .
python3 -m pkinv.sectors.ippu.run_ippu
python3 -m unittest discover tests
python3 -m pkinv.report.figures
python3 scripts/sync_figures.py
```

The run generates the sector's headline tables—`ippu_totals_2024.csv` (the result by category in both common-GWP vintages), `ippu_by_stream_2024.csv` (category, subitem and gas in gigagrams with each row's uncertainty, activity year and period kind, evidence class, temporal and spatial method and source), `ippu_provincial_2024.csv` (the province-level AR5 total, each row carrying the allocation status that produced it) and `ippu_monthly_2024.csv` (the twelve-month layer with each month's share, the emissions it implies and the basis of the shape)—together with the evidence tables below. Each is named, because a reader reproducing the run is otherwise told what it produces without being able to tell which file is which; every quantity this paper prints from one of them is checked against it by the verification registry.

`ippu_cement_material_balance_2024.csv` the clinker material balance: domestic production, trade and stock terms against inferred clinker.

`ippu_cement_producer_census_2024.csv` the 16-company producer census, each entity's reported clinker quantity (Section 3).

`ippu_company_clinker_2024.csv` company-level clinker flows for the same 16 entities.

`ippu_company_clinker_coverage_2024.csv` the company-against-census coverage reconciliation—identical universe and period, zero stock change.

`ippu_cement_valid_structural_2024.csv` the valid trade-corrected structural models: scenario, cement source, clinker fraction and resulting CO_2.

`ippu_cement_tier_comparison_2024.csv` the same estimate by method and coverage, the producer census as reference.

`ippu_cement_method_benchmarks_2024.csv` the two retained no-trade benchmarks, marked screening-only, isolating the bias from omitting observed clinker trade.

`ippu_cement_conditional_intervals_2024.csv` conditional Monte Carlo intervals per configuration, with the deterministic value, the quantiles, the draw count and the seed.

`ippu_cement_matched_calendar_2022.csv` the matched calendar-2022 reconstruction, inferring clinker production from that year's own activity and trade.

`ippu_cement_activity_reconciliation.csv` a reported cement quantity against the period it actually belongs to, retained so a superseded period tag cannot re-enter as a structural alternative.

`ippu_ammonia_activity_reconciliation.csv` ammonia activity by pathway against the official filing, with each pathway's independence from it recorded.

`ippu_fertilizer_gas_cross_source_2024.csv` the fertiliser gas cross-source check: NFDC plant meters against the gas ledger, with the feed and fuel shares and their agreement.

`ippu_chemical_plant_activity.csv` the chemical-plant evidence table (Section 9 states what it does not cover).

`ippu_activity_controls_2024.csv` the activity controls, one row per IPCC code and product, each with its source reference, coverage and data tier.

`ippu_facility_registry.csv` the facility registry and `ippu_facility_coverage.csv` its geocoding coverage by district—declared, geocoded and ungeocoded counts (Section 3).

The run also writes the CRT activity–factor bridge, the full completeness matrix, nitric-acid evidence, HFC data specifica-

tions, the source manifest, the CRT cell extraction, the quality-control report and the run manifest.

The run manifest records every input hash, output name, reference result and structural range. The CRT cell audit additionally records the source ZIP and workbook hashes, worksheet, cell, raw value, formula and cell-value hash. The generated national total contains only the independently estimated 2.A.1 process CO_2 component.

Appendix B. Completeness and data requirements

Each detailed category is documented in `ippu_completeness_2024.csv` by IPCC code, gas, notation key, activity and method status, public source, available years, source role, materiality and remaining data requirement—the table this section has always described and never named. The gas split behind the fertiliser feedstock is published beside it in `ippu_fertilizer_gas_monthly_2024.csv`, which carries feed and fuel volumes in million cubic feet by month with each month's share of the year and the feedstock share it implies. The highest-priority data needs are:

1. APCMA and cement producers: clinker production, purchases, sales, trade and stocks;
2. NFDC and fertiliser producers: ammonia and 100%-nitric-acid output, feedstock, recovered CO_2, process technology and N_2O abatement;
3. National Ozone Unit and Customs: annual chemical/application imports, exports, destruction and equipment charge for HFCs;
4. PBS and industrial producers: marketed and captive lime, glass tonnes/type/cullet, carbonate inputs, steel route and carbon inputs, and lubricant consumption;
5. NTDC, distribution companies and major equipment owners: SF_6 equipment, purchases, refills, recovery and retirement.

Appendix C. Source hierarchy

Methods and cement factors are from IPCC Volume 3 (IPCC, 2006). FY2023–24 cement activity is from PBS QIM (Pakistan Bureau of Statistics, 2024) and is corroborated by the SBP fiscal series (State Bank of Pakistan, 2024); that cross-check corrects the former Pocket Book period tag. FY2023–24 clinker trade is from UN Comtrade (United Nations Statistics Division, 2024), and calendar-2022 trade is cross-checked through WITS/UN Comtrade (World Bank, 2022); company clinker sources are recorded row by row in the producer ledger. The official comparison is Pakistan's 2022 CRT (Government of Pakistan, 2026), and HFC consumption data are from the Ozone Secretariat country profile (Ozone Secretariat, 2026).